

4.3

SUSTAINABILITY

It is defined as *“meeting our own needs without compromising the ability of future generations to meet their own needs”*.

4.3.1 Need of sustainability

1. Sustainability is key to preserving our planet.
2. Sustainability helps reduce pollution and conserve resources.
3. Sustainability creates jobs and stimulates the economy.
4. Sustainability improves public health.

5. It protects biodiversity.
6. It protects the natural environment.
7. It is the choice of non-toxic materials.
8. It reduces and reuses the resources.
9. It minimizes waste.
10. It is used for life-cycle analysis.

4.3.2 Concept (or) Approaches (or) Significance of Sustainability

To build up the sustainability development, the following approaches (or) methods are proposed.

1. Developing appropriate technology: It is the one, which is locally adaptable, eco-friendly, resource-efficient and culturally suitable. It uses local labours, less resources, and produces minimum waste.

2. Reduce, Reuse, Recycle (3-R) approach: It insists optimum use of natural resources, using it again and again instead of throwing it on the waste land (or) water and recycling the material into further products. It reduces pressure on our natural resources and reduces waste generation and pollution.

3. Providing environmental education and awareness: By providing environmental education and awareness, the thinking and attitude of people towards our earth and the environment can be changed.

4. Consumption of Renewable Resources: In order to attain sustainability, it is very important to consume the natural resources in such a way that the consumption should not exceed regeneration capacity.

5. Conservation of non renewable resources: Non-renewable resources should be conserved by recycling and reusing.

6. Population Control: By controlling population growth, we can make very good sustainability development.

4.3.3 Economic and Social Challenges of Sustainability

1. Economic sustainability

It refers to the organisation's ability to manage its resources and responsibly generate profits in the long term.

Examples

1. A company uniliver

It has followed a strategy to achieve a balance between sustainability and the company's economic performance. So, it implemented several measures like increasing package recycling, promoting the use of recycled materials and responsible consumption awareness campaigns.

2. A company suez

It has reduced its emissions, related to electricity consumption, by 95% by using renewable energy and conservation of natural habitats.

Economic challenges

- (i) High rates of unemployment (or) under employment.
- (ii) High rates of poverty and low growth.
- (iii) Increasing inequality, with many not being included in the growth process.
- (iv) Disruption of major economic activities due to pandemic situation like tourism.

- (v) Volatile growth dependent on one source.
- (vi) Low productivity due to poor human capital development.
- (vii) Skills mismatch between skills you have and the jobs you want to create.
- (viii) Lack of quality jobs.
- (ix) Macroeconomic instability and recurrent balance of payments shocks.

2. Social sustainability

It refers to strengthening the cohesion and stability of specific social groups.

Examples

1. A Company CEMEX

It is working to contribute to the social development of communities. Thus, it offers decent housing through self-building programmes and loans with favourable access conditions.

2. A Gigante group

It contributes funds and resources to a range of social causes like school materials for collaborators and grants to improve visual health.

Social Challenges

Though social impact, social sustainability challenges, issues are not easily measurable, they are easier to identify. Social sustainability performance challenges include

- (i) Human rights.
- (ii) Fair labour practices.
- (iii) Living conditions.
- (iv) Health and safety.

- (v) Wellness, diversity and equity.
- (vi) Work-life balance.
- (vii) Empowerment.
- (viii) Community engagement.

4.3.4 Aspects of sustainability

There are 4 aspects of sustainability, of them environmental sustainability is the fundamental and important aspect.

1. Environmental aspect

Environmental aspect acknowledges the need to enhance and maintain the biophysical systems that sustain all the life on earth. It includes the structure and function of natural ecosystems and the interactions between them and people and calls for guardianship (or) kaitiakitanga of our environment.

2. Social aspect

Social aspect acknowledges the need for equity

- (i) within and between generations, and
- (ii) within and between ethnic and social groups.

It is inclusive of people's mental and physical well-being and the cohesion of their communities based on a fair distribution of resources.

3. Cultural aspect

Cultural aspect acknowledges the need to nourish and share attitudes and values that represent diverse world views and the political need for all people to express their views freely and to participate in decision making. Addressing these needs can build resilience for the future.

4. Economic aspect

Economic aspect acknowledges the interactions of humans with the natural environment in using resources to create goods and services which add value to their lives. It acknowledges the resource use and waste disposal must occur within the capacity of our planet. It encourages a fair trading system that equitably distributes benefits and costs. It further encourages innovation and creativity in developments that lead to a sustainable future.

Relationship between these aspects

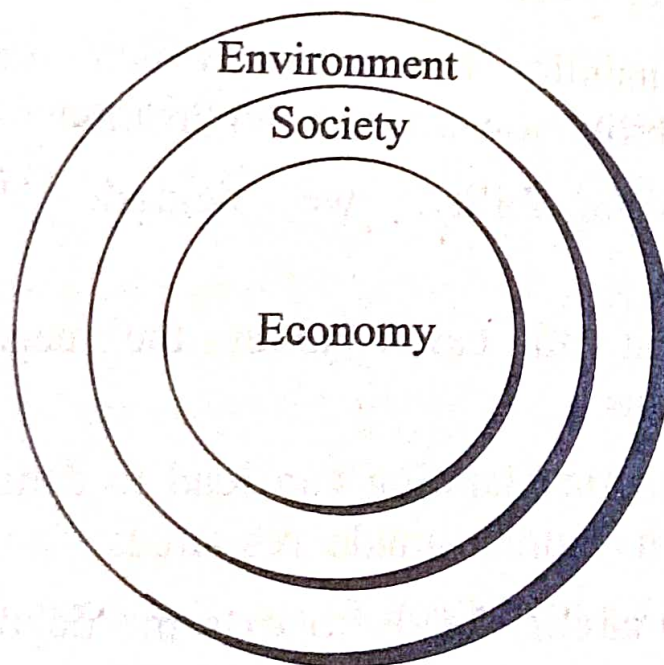


Fig. 4.2: Strong sustainability aspects

This model shows how our economy is a subset of our society. It also shows that everything in our economy and everything in our society entirely dependent on our environment.

This relationship means that any impact (or) change to our environment will impact on society and the economy. Therefore any sustainability related issue must be considered holistically and recognise their interdependence.

4.4

FROM UNSUSTAINABILITY TO SUSTAINABILITY**4.4.1 Unsustainability**

Our ancestors have left a lot of resources for us. They used their resources sensibly and not for their greed. But we are exploiting limited resources. Instead of using it for our needs, we are exploiting it for our greeds.

Characteristics of Unsustainability

- (i) Unsustainability is one in which we forget our responsibility towards the environment.
- (ii) In unsustainability, we degrade the available resources.
- (iii) It not at all cares about the needs of future generations.
- (vi) Unsystematic planning can lead to damage to natural as well as human-made resources.
- (v) Unsustainability leads to extreme degradation of the environment as well as the living organism.

Causes for unsustainability

- (i) Developing countries are responsible for the degradation.
- (ii) The rate of increase of National pollution. In this regard, developed countries contribute much more than developing countries.
- (iii) Raising population.
- (iv) People should look at environment as not only reserve of man but of all living organism.

- (v) People built so many buildings, roads and dams for a luxurious life-style. We neglected the needs of animals and destroy their habitats.
- (vi) We extract a lot of material from the lithosphere than what we need.
- (vii) We create a lot of chemical compounds, which break down in the environment and becomes problematic.
- (viii) We cut trees at a faster rate than they can grow, which causes global warming.
- (ix) Purchasing and using polythene bags contribute to growth of pollution.

4.4.2 Characteristics of Sustainability

1. It reduces emission of greenhouse gases, which will reduce global warming and helps in preserving the environment.
2. It uses natural and biodegradable materials for reducing the impact on the environment.
3. It emphasis on using renewable energy sources such as wind and solar energy.
4. It follows non-polluting construction practices.
5. It protects the natural habitats.
6. It improves the quality of human life.
7. It minimises the depletion of natural resources.
8. It teaches us to respect and care for all the life forms.
9. It makes arrangements, so that the future generations are able to meet their own demands.

Table 4.1 Differences between sustainability and unsustainability

S.No.	Sustainability	Unsustainability
1.	Prevention of natural resources.	Exploitation of natural resources is faster than the planet can handle and replenish.
2.	Long-term economic growth without negatively impacting our environment (or) society.	Not quantifying ecosystem services and increased vulnerability to crises.
3.	Equality, diversity, social cohesion and democracy.	Labour laws, human rights, gender inequality and poor treatment of indigenous people.

4.5

MILLENNIUM DEVELOPMENT GOALS

The Millennium Development Goals (MDGs) were 8 international development goals.

1. To eradicate extreme poverty and hunger.
2. To achieve universal primary education.
3. To promote gender equality and empower women.
4. To reduce child mortality.
5. To improve maternal health.
6. To combat HIV/AIDS, malaria, and other diseases.
7. To ensure environmental sustainability.
8. To develop a global partnership for development.

4.6

SUSTAINABILITY PROTOCOLS

Sustainability protocols are sustainability standards and certifications. These are voluntary guidelines used by producers, manufacturers, traders, retailers and service providers to demonstrate their commitment to good environmental, social, ethical and food safety practices.

There are over 400 such standards across the world. The sustainability protocols listed below are important because they build awareness and policy support, create clear guideline and goals. They have third-party verification and maintain consistency within a portfolio while developing Green economy.

Few sustainability protocols

1. LEED - Leadership Energy & Environmental Design
 2. WELL -
 3. Fitwel
 4. Living building challenge
 5. BREAM
 6. Passive house
 7. National Green Building Standard
 8. Built green
 9. Evergreen sustainable development standard (ESDS)
- (a) Introduction of eco-labels and standards for organic food and other food products.
 - (b) Triple bottom line, it includes a set of practices (or) criteria for how a crop should be sustainably grown (or) a resource should be ethically harvested.
 - (c) It includes, responsible fishing practices that do not endanger marine biodiversity (or) respect for human

- rights and the payment of fair wages on a coffee (or) tea plantation.
- (d) Sustainability protocols are accompanied by a verification process (certification) to evaluate that an enterprise complies with a standard as well as a traceability process for certified products to be used along the supply chain, often resulting in a consumer-facing label.
- (e) It also focus on capacity building and working with partners and other organizations to support small holders (or) disadvantages producers to make the social and environmental improvements needed to meet the standard.

4.7

SUSTAINABLE DEVELOPMENT

Definition

Sustainable development is the development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

4.7.1 Aim of true sustainable development

It aims at optimum use of natural resources with high degree of sustainability, minimum wastage, least generation of toxic by-products and maximum productivity.

1. **Inter - generational equity:** It states that we should hand over a safe, healthy and resourceful environment to our future generations.
2. **Intra - generational equity:** It states that the technological development of rich countries should support the economic growth of the poor countries and help in narrowing the wealth gap and lead to sustainability.

3. Place more emphasis on pollution prevention and waste reduction.
4. Recycle and reuse as many of our waste products and resources possible.
5. Make more goods that last longer and easy to use, recycle and repair.
6. Depends on renewable sources of energy (sun, wind, water, bio-mass).
7. Sustain earth's biodiversity.
8. Earth degrading activities should be discouraged.
9. Reduce poverty and rate of population growth.
10. Don't use high quality energy to do a job.

4.7.2 Goals of Sustainable Development

There are a total of 17 goals

1. End poverty everywhere, in all its forms.
2. End hunger, achieve food security and improved nutrition.
3. Ensuring good health and promote the well-being of all age groups.
4. Ensuring inclusive and equitable quality education.
5. Achieving gender quality and empowering all women and children.
6. Ensuring the availability and sustainable management of water and sanitation for all.
7. Ensuring access to affordable, reliable, clean and modern energy for all.
8. Promoting sustainable economic growth.
9. Building resilient infrastructure and promoting sustainable industrialisation.

10. Reducing inequality within and among countries.
11. Making cities and human settlements inclusive, safe, resilient and sustainable.
12. Ensuring sustainable consumption and production patterns.
13. Taking urgent action to combat climate change and its impacts.
14. Conserving and sustainably using the oceans, seas and marine resources for sustainable development.
15. Protecting, restoring and promoting the sustainable use of terrestrial ecosystems.
16. Promoting peaceful societies for sustainable development.
17. Strengthening the means of implementation and revitalising the global partnership for sustainable development.

4.7.3 Sustainable development targets

1. Eradicate extreme poverty for all the people everywhere.
2. Reduce at least by half the proportion of men, women and children of all ages living in poverty.
3. Implement nationally appropriate social protection systems and measures for all including floors.
4. Ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, natural resources and new technology and financial services.
5. Build the resilience of the poor and those in vulnerable situations and reduce their exposure to climate related events and disasters.

6. Ensure significant mobilization of resources from a variety of sources from developed countries to less developed countries.
7. End hunger and ensure access by all people, in particular the poor people and people in vulnerable situations.
8. End all forms of malnutrition.
9. Double the agricultural productivity and income of small scale food producers.
10. Ensure sustainable food production systems and implement resilient agricultural practices.
11. Maintain the genetic diversity of seeds, cultivated plants and farmed animals.
12. Increase investment in rural infrastructure, agricultural research and technological development to enhance agricultural productive capacity.
13. Correct and prevent trade restrictions and distortions in world agricultural markets.
14. Adopt measures to ensure the proper functioning of food commodity markets.
15. Reduce the global maternal mortality ratio to less than 70 per 1,00,000 live births.
16. End preventable deaths of new borns and children under 5 years of age.
17. End the epidemics of AIDS, tuberculosis, malaria, hepatitis, water-borne diseases.
18. Reduce by one third premature mortality from non-communicable diseases through prevention and treatment.
19. Strengthen the prevention and treatment of substance abuse like narcotic drug abuse and harmful use of alcohol.

20. Halve the number of global death and injuries from road accidents.
21. Ensure universal access to sexual and reproductive healthcare services like family planning.
22. Achieve universal health coverage including financial risk protection.
23. Reduce the number of deaths and illnesses from hazardous chemicals and air, water, soil pollution.

4.7.4 Sustainable development indicators

1. Proportion of the population living below the **international poverty line** by sex, age, employment status.
2. Proportion of the population living below the **national poverty line** by sex and age.
3. Proportion of population covered by social protection floors.
4. Proportion of total adult population with secure tenure rights to land with legally recognized documentation.
5. Proportion of local governments that adopt and implement local disaster risk reduction strategies in line with national disaster risk reduction strategies.
6. Proportion of total government spending on essential services.
7. Prevalence of under nourishment.
8. Prevalence of stunting and mal-nutrition among children and 5 years of age and prevalence of anaemia in women aged 15 to 49 years.
9. Average income of small-scale food producers.
10. Proportion of agricultural area under productive and sustainable agricultural practices.

11. Proportion of local breeds classified as being at risk of extinction.
12. Total official flows to the agricultural sector.
13. Agricultural export subsidies.
14. Indicator of food price anomalies.
15. Maternal mortality ratio.
16. Under - 5 mortality rate.
17. Hepatitis B (or) Tuberculosis incidence per 100,000 population. Malaria incidence (or) HIV infections per 1000 population.
18. Suicide mortality rate.
19. Alcohol per capital consumption within a calendar year in litres of pure alcohol.
20. Death rate due to road traffic injuries.
21. Proportion of women of reproductive age, who have their need for family planning.
22. Coverage of essential health services.
23. Mortality rate due to unsafe water, sanitation and lack of hygiene.

4.7.5 Intervention areas of sustainable development goals

Implementation of sustainable development goals are some what difficult because of the following interventions.

1. Climate change.
2. Use of natural resources.
3. Waste production.
4. Water pollution.
5. Deforestation.

6. Over fishing.
7. Poverty.
8. Ocean acidification.
9. Air pollution.

4.8**CLIMATE CHANGE**

Climate change refers to the long-term shifts in temperatures and weather pattern. These changes may be natural (through variations in solar cycle) (or) artificial (human activities like burning of fossil fuels like coal, oil and gases).

4.8.1 Causes of climate change

1. Presence of green house gases in the atmosphere increases the global temperature.
2. Depletion of ozone layer also increases the global temperature.
3. Uneven distribution of rainfall.
4. Rotation of earth on its axis.
5. Seasonal changes.

4.8.2 Effect (or) Issues of climate change

1. Even small changes in climatic conditions may disturb agriculture that would lead to migration of animals including humans.
2. Climate change may upset the hydrological cycle, results in floods and droughts in different regions of the world.
3. Global pattern of winds and ocean currents also gets disturbed by climate.

4. More frequent and intense drought.
5. Sea levels gets raised.
6. Melting glaciers and warming oceans can directly harm animals and destroy the places they live.
7. Higher temperature increases heat related illnesses and make working outdoors more difficult.
8. If conditions are hot wild fires start more easily and spread more rapidly.
9. As green house concentration increases, global surface temperature also increases.

4.8.3 Possible solutions to climate change

1. Burning of fossil fuels like coal, oil and gas must be avoided.
2. Renewable energy sources like solar, wind, tidal and geothermal power must be used instead of fossil fuels.
3. Reducing petrol and diesel vehicles, planes and ships and switching to electric vehicles stop climate change.
4. Heat our homes in a green way, by insulating walls and roofs and switching away from oil (or) gas boilers to heat pumps.
5. The best way for individuals to help stop climate change is by reducing their meat and dairy consumption (or) by going fully vegetarian.
6. Planting more trees in the right place will absorb more carbon from the emissions.
7. Protect forests, which fight against climate change.
8. Oceans also absorb large amounts of CO_2 from the atmosphere, which helps to keep our climate stable.

9. Reducing overall consumption in more wealthy countries can help put less strain on the planet.
10. Avoid of using plastics.
11. Reduce carbon pollution to avoid the worst consequence of climate change.
12. Provide financial support to developing countries, so people and nature can successfully adapt.

4.9**CASE STUDIES****4.9.1 Climate change on Chennai, East Coast Road (ECR) and Old Mahabalipuram Road (OMR)**

ECR and OMR are the today's landmark will receive the highest climate impact. Due to Chennai's land use patterns, population stress and abuse of natural resources, climate has changed drastically in the last few years. Climate induced impacts like drought, floods, heavy rains and winds are becoming increasingly evident in the city.

On one end of the continuum is summer water crises and on the other end is monsoonal flood disasters.

Reason for flood in Chennai

Increasing population and building / Land use pattern have changed the natural hydrology of the city. The water which needs to naturally drain into the sea via, the regional watershed are now blocked by buildings and artificial man-made structures. These are all reason for flood in Chennai city.

Remedy

Researchers predicts that Chennai's climate is more unpredictable and aggressive. Climate change will affect

people and the environment. People need to take action to reduce the emission of green house gases that is the root cause for climate change.

4.9.2 Climate change on Chennai, Ennore

Ennore thermal power station (ETPS), a 660 MW coal-fired thermal power plant in Ennore at north Chennai, CPCL's oil refinery, Madras Fertilizers Ltd, Tamil Nadu Petroproducts Ltd and Madras Petrochemical Ltd. These six factories were, operating in violation of prescribed air pollution norms for nearly 60%, responsible for drastic climate change.

More than 56 lakh tonnes of coal ash is spread over the river bed with flyash deposits ranging in depth from 1 ft to 8 ft.

Issues (or) effects

1. Ground-level particulate matter pollution due to emissions, from above said 6 factories at around Ennore, exceeded the carrying capacity of the area in the vicinity of the plant.
2. Ground-level SO_2 and NO_2 pollution exceeded the carrying capacity of the area.
3. In Ennore, children and women are particularly affected and gynecological problems were particularly reported by respondents.
4. Natural drainage pattern have been considerably altered due to fly ash pond construction and ash contamination. This will have an impact on local hydrology and flooding.
5. Climate change manifests in the form of rising sea levels, increased heat stress, intense rain events and

droughts and ocean desertification due to warming seas.

6. More than 120 million litres of hot waste water is discharged daily, from Ennore power plant, into the Ocean. Such discharge in an already warming Ocean will create localised marine deserts.

4.10 CARBON CREDIT

4.10.1 Definition

A carbon credit is a tradable permit (or) certificate that represents the right to emit a set amount of CO_2 (or) 1 tone of CO_2 (or) the equivalent amount of green house gas.

4.10.2 Concept

Kyoto protocol is an international agreement that aims to manage and reduce carbon dioxide emissions and green house gases. Kyoto protocol introduced the concept of carbon credits.

According to this, a country should reduce carbon emissions in the atmosphere.

- (i) A carbon credit is a tradable certificate that allows its holder to emit green house gases.
- (ii) One carbon credit is equal to one ton of carbon dioxide.
- (iii) Countries need to reduce their emissions by 5.2% compared to the numbers recorded.
- (iv) Countries and companies need to be designed to reduce carbon emissions without the need to buy credits.

- (v) Less the purchase, less will be the carbon release into the atmosphere.

4.10.3 Types of carbon credits

There are two types of carbon credits.

1. Voluntary emissions reduction (VER)

It is a carbon offset that is exchanged in the over-the-counter (or) voluntary market for credits.

2. Certified emissions reduction (CER)

It relies on emission credits created through a regulatory frame work with the purpose of off-setting a project's emissions.

4.10.4 How to get carbon credit

Carbon credits and carbon markets are a component of a national and international attempts to mitigate the growth in concentrations of green house gases (GHGs). One carbon credit is equal to one ton of CO₂ (or) CO₂ equivalent gases.

There are many companies, that sell carbon credits to commercial and individual customers who are interested in lowering their carbon foot print. Buyers and sellers can also use an exchange platform to trade, which is like a stock exchange for carbon credits.

4.10.5 Advantages and Disadvantages of Carbon Credits

Advantages of Carbon Credit

1. Each carbon credit corresponds to one ton of carbon that was not emitted into the atmosphere.

2. The company that does not have an alternative to reduce its emission finds an advantage in the purchase of this credit.
3. The purchase of carbon credits by companies that are environmentally conscious is favorable for their image.
4. Sometimes the company behaves sustainably, but it cannot stop emitting some amount of carbon. In this case, the purchase of credit shows how much the company cares about encouraging sustainable activities.
5. It enables companies to support decarbonization beyond their own carbon footprint.

Disadvantages of Carbon Credit

1. Some institutions and countries can accommodate themselves in the exchange market to continue emitting their greenhouse gases.
2. Companies do not invest in action to avoid emissions because they are able to buy unlimited credits.
3. The reduction of 1 ton of carbon that is 1 credit, will never be enough.
4. It is an alternative for emergency needs, not to rest on the fact that the other has saved.
5. Main focus of carbon credit is to reduce green house gas emissions, but it is not possible to stop the negative impacts caused by the globe warming.

4.11

CARBON FOOTPRINT**Definition**

It is the total amount of green house gases (including CO_2 and CH_4) that are generated (emitted) by our direct and indirect activities.

Individual carbon footprint

It is the sum total of their direct and indirect carbon emissions over the course of a year.

- i.e., Smaller your carbon footprint : better for the future
Bigger your carbon footprint : Have bigger negative impact in environment

The average carbon footprint for a person in united state is 16 tons. Globally, the average is closer to 4 tones. To avoid 2°C rise in global temperatures, the average global carbon footprint per year needs to drop under 2 tons by 2050.

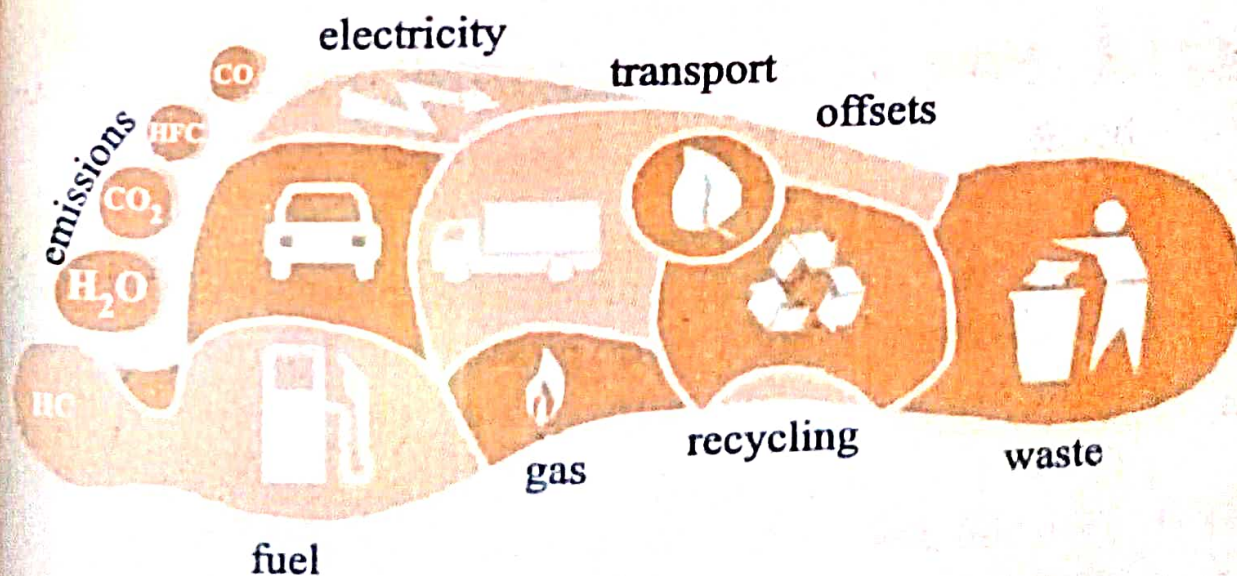


Fig. 4.3 Carbon Footprint

4.11.1 Sources of carbon footprint

1. Climate change.
2. Natural process like volcanos.
3. Green house gases emitted from human activities.
4. Pollution released by human beings doing human things.
5. Transportation accounted for about 28% of total country.
6. Electricity generation accounted for about 28%.
7. Industrial activities 22%.
8. Heating and cooling in homes and businesses contribute 11%.

4.11.2 Causes of a carbon footprint

The major contributors to carbon footprints are

- (i) food: (especially meat (beef))
- (ii) consumption
- (iii) transportation
- (iv) house hold energy

4.11.3 How to lower (control) carbon footprint (or) 15 ways to reduce your carbon footprint

Lowering individual carbon footprint from 16 tons to 2 tons does not happen over night. But, by making small changes in our action we can reduce carbon footprint.

Once you understand where your emission comes from, you can take steps to reduce your impact.

1. Calculate your carbon footprint.
2. Drive less.
3. Switch to an electric (or) hybrid car.

4. Travel smart.
5. Switch to renewable energy.
6. Consider solar panels.
7. Make your home more efficient.
8. Turn your thermostat just 2 degrees cooler in winter and 2 degrees warmer in summer.
9. Get energy efficient appliances.
10. Unplug electrical devices when not in use.
11. Buy locally - sourced food.
12. Start a home garden.
13. Eat less meat.
14. Don't waste water.
15. Reduce, reuse and recycle.